ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Digital Logic Design

هذا العمل صدقة جارية عن أرواح موتى وشهداء المسلمين عامة وشهداء فلسطين خاصة
اللهم إنهم في ذمتك وحبل جوارك، فَقِهِم من فتنة القبر وعذاب النار، وأنت أهل الوفاء والحق، اللهم اغفر لهم وارحمهم، فإنك أنت الغفور الرحيم
اللهم اغفر لهم وارحمهم واعف عنهم وعافهم، وأكرم نزلهم ووسع مدخلهم، واغسلهم بالماء والثلج والبرد، ونقهم من الخطايا كما ينقى الثوب الأبيض من الدنس، وأبدلهم داراً خيراً من دارهم وأهلاً خيراً من أهلهم وأزواجاً خيراً من أزواجهم، وقهم فتنة القبر وعذاب النار
اللهم إنهم عبيدك وابناء عبيدك وابناء إماءك كانوا يشهدون أن لا إله إلا أنت، وأن محمداً عبدك ورسولك، وأنت أعلم بهم، اللهم إن كانوا محسنين فزدهم في إحسانهم، وإن كانوا مسيئين فتجاوز عن سيئاتهم، اللهم لا تحرمنا أجرهم ولا تفتنا بعدهم

اللهم سدد رمي المقاومة
اللهم لا ترفع لليهود راية ولا تحقق لهم غاية واجعلهم لمن خلفهم عبرة وآية

قال تعالى:{﴿ سُبْحَانَ الَّذِي أَسْرَى بِعَبْدِهِ لَيلاً مِنَ الْمَسْجِدِ الْحَرَامِ إِلَى الْمَسْجِدِ الأَقْصَا الَّذِي بَارَكْنَا حَوْلَهُ لِنُرِيَهُ مِنْ آيَاتِنَا إِنَّهُ هُوَ السَّمِيعُ الْبَصِيرُ * وَآتَيْنَا مُوسَى الْكِتَابَ وَجَعَلْنَاهُ هُدىً لِبَنِي إِسْرَائِيلَ أَلاَّ تَتَّخِذُوا مِنْ دُونِي وَكِيلاً * ذُرِّيَّةَ مَنْ حَمَلْنَا مَعَ نُوحٍ إِنَّهُ كَانَ عَبْداً شَكُوراً * وَقَضَيْنَا إِلَى بَنِي إِسْرَائِيلَ فِي الْكِتَابِ لَتُفْسِدُنَّ فِي الأَرْضِ مَرَتَيْنِ وَلَتَعْلُنَّ عُلُواً كَبِيراً * فَإِذَا جَاءَ وَعْدُ أُولاهُمَا بَعَثْنَا عَلَيْكُمْ عِباداً لَنَا أُولي بَأْسٍ شَدِيدٍ فَجَاسُوا خِلالَ الدِّيارِ وَكَانَ وَعْداً مَفْعُولاً * ثُمَّ رَدَدْنَا لَكُمُ الْكَرَّةَ عَلَيْهِمْ وَأَمْدَدْنَاكُمْ بِأَمْوَالٍ وَبَنِينَ وَجَعَلْنَاكُمْ أَكْثَرَ نَفِيراً * إِنْ أَحْسَنْتُمْ أَحْسَنْتُمْ لِأَنْفُسِكُمْ وَإِنْ أَسَأْتُمْ فَلَهَا فَإِذَا جَاءَ وَعْدُ الآخِرَةِ لِيَسُوؤا وُجُوهَكُمْ وَلِيَدْخُلُوا الْمَسْجِدَ كَمَا دَخَلُوهُ أَوَّلَ مَرَّةٍ وَلِيُتَبِّرُوا مَا عَلَوْا تَتْبِيراً * عَسَى رَبُّكُمْ أَنْ يَرْحَمَكُمْ وَإِنْ عُدْتُمْ عُدْنَا وَجَعَلْنَا جَهَنَّمَ لِلْكَافِرِينَ حَصِيراً ﴾} سورة الإسراء

Eng. Mohannad Akram Mohammed Al Manasra, BAU-FET Electrical Power Engineering Graduate
بسواعدنا نبنيها

والكمال لله وحده لهذا من الممكن أن تتواجد بعض الأخطاء في الحلول لذلك نرجو منكم الاستفسار عن الأسئلة التي تريدون السؤال عنها بالتواصل مع مُنشأ هذا العمل

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From the truth table below, determine the standard SOP expression:

Inputs			Output
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Select one:

- a. $X = \bar{A}\,\bar{B}\,C + \bar{A}\,B\,C + A\,B\,\bar{C}$
- b. $X = \bar{A}\,\bar{B}\,\bar{C} + A\,B\,C + A\,\bar{B}\,C$
- c. $X = A\,B\,C + A\,B\,C + A\,B\,C$
- d. $X = A\,\bar{B}\,C + \bar{A}\,B\,C + A\,B\,\bar{C}$

How many (6) you have to add to find the final answer using BCD adding for the operation :

77 + 88

Select one:

- a. twice
- b. 1 time
- c. 3 times
- d. no need to add 6

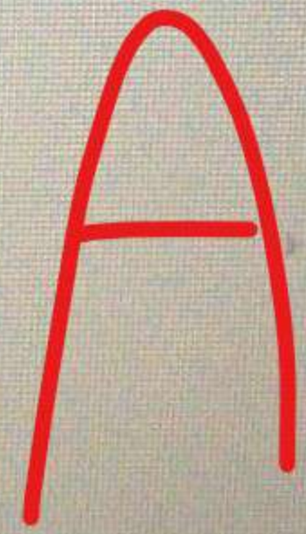
B

Minimize the function F and express it in SOP:

$$F('A,B,C,D) = \sum (0,2,4,6,7,10,3,14,15)$$

Select one:

- a. $f = C\overline{D} + BC + \overline{A}C + \overline{\overline{A}D}$
- b. $f = C\overline{D} + B\overline{C} + AC + A\overline{D}$
- c. $f = C\overline{D} + \overline{\overline{B}C} + \overline{A}C + AD$
- d. $f = CD + BC + AC + AD$

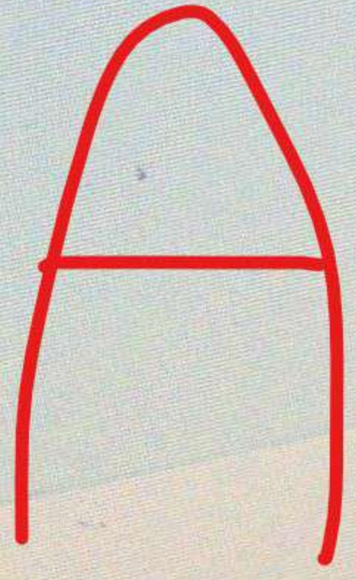


DESIGN-3 ▶ General ▶ الامتحان النصفي

The binary number **- 1101.011** can be represented in **sign and magnitude** form as :

Select one:

- a. **11101.011**
- b. **01101.011**
- c. **0101.011**
- d. **1101.1100**



GIC DE

SIGN-3 ▶ General ▶ ...

The result of subtraction $(CDE)_{16} - (477)_8$ in octal is:

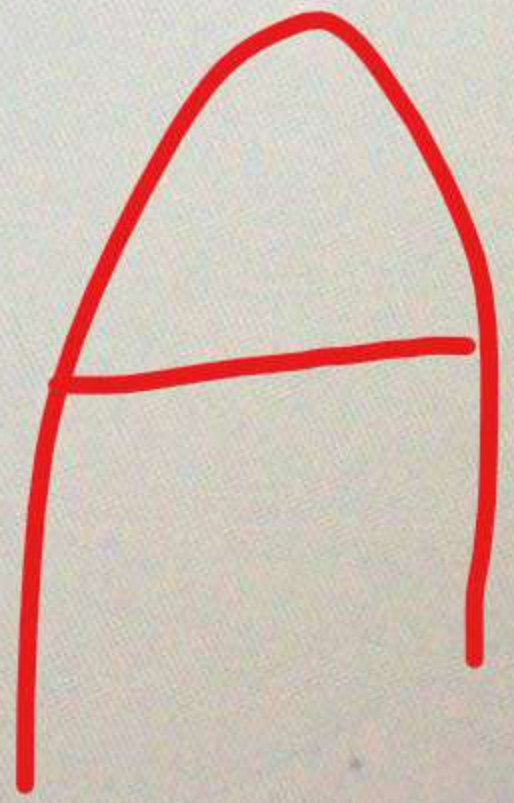
Select one:

a. $(5757)_{octal}$

b. $(5547)_{octal}$

c. $(5637)_{octal}$

d. $(5736)_{octal}$



The following boolean function $(X + Y + XY)(X + Z)$ can be simplified to :

Select one:

- a. Y+Z
- b. XYZ
- c. X+YZ
- d. X



The POS function which equivalent to SOP function $F(A,B,C)=\sum m(1,3,4,5,7)$ is?

Select one:

- a. $F(A,B,C)=\prod M(0,1,3,4,5)$
- b. $F(A,B,C)=(\overline{A}+\overline{B}).(\overline{B}+C).(A+B+C)$
- c. $F(A,B,C)=(\overline{A}+\overline{B}+C).(A+B+C).(A+\overline{B}+C)$
- d. $F(A,B,C)=\prod M(0,2,6,7)$

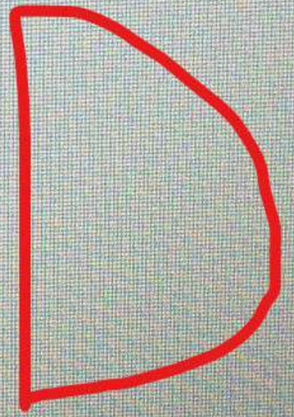
C

Simplify the following boolean function :

$$AB + A\overline{C} + C + AD + A\overline{B}C + ABC$$

Select one:

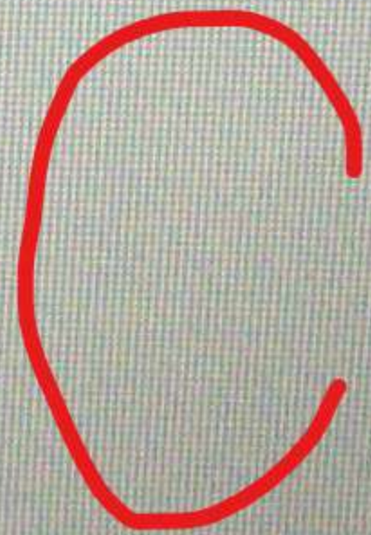
- a. A+B+C
- b. A'+C
- c. A' + D'
- d. A+C



The result of -77 + -33 in 2's complement is :

Select one:

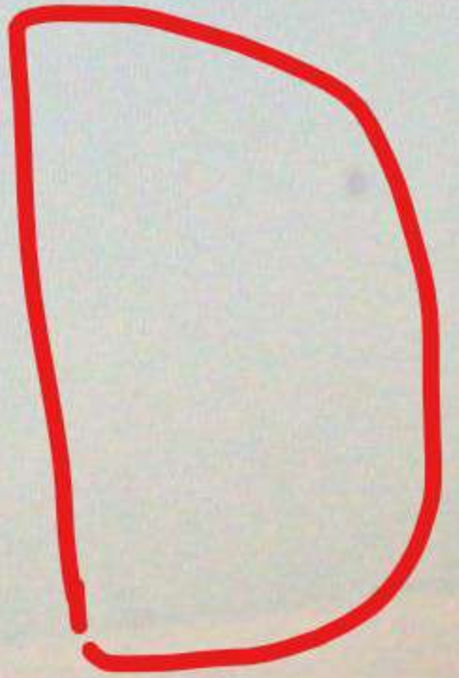
- a. 10010001
- b. 10010010
- c. 01101110
- d. 10010011



The binary code $(11110101)_2$ is equivalent to ($)_{gray}$

Select one:

- a. 11001111
- b. 11110001
- c. 10101010
- d. 10001111



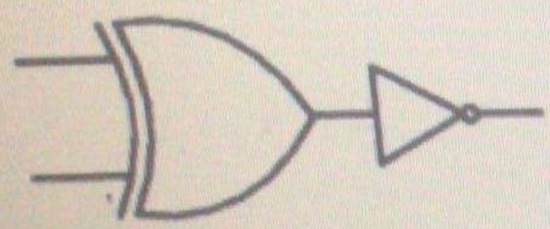
The OCTAL number equivalent to $(7BA.CC)_{16}$ is :

Select one:

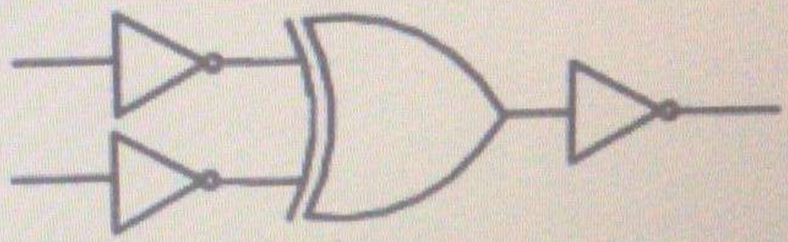
- a. $(3672.314)_8$
- b. $(3673.36)_8$
- c. $(3672.63)_8$
- d. $(3772.67)_8$

Which one of the following circuits is **NOT** equivalent to a 2-input XNOR (exclusive NOR) gate?

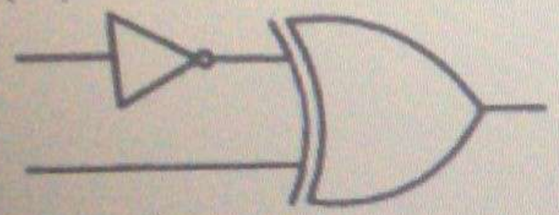
(A)



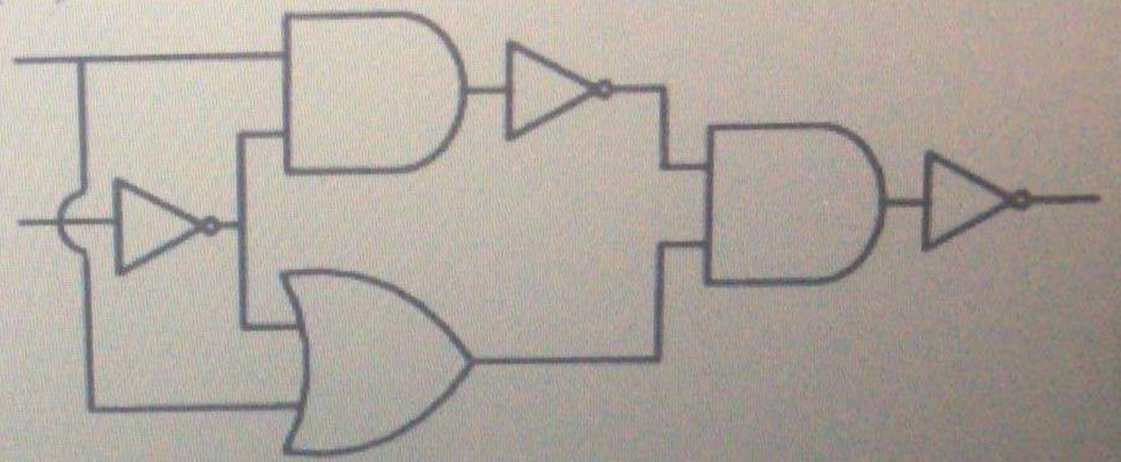
(B)



(C)



(D)



Select one:

- a. C
- b. A
- c. D
- d. B

A 6x64 line decoder can be built using:

Select one:

- a. six 2x4 line decoders only
- b. nine 2x4 line decoders only
- c. seven 3x8 line decoders only
- d. nine 3x8 line decoders only



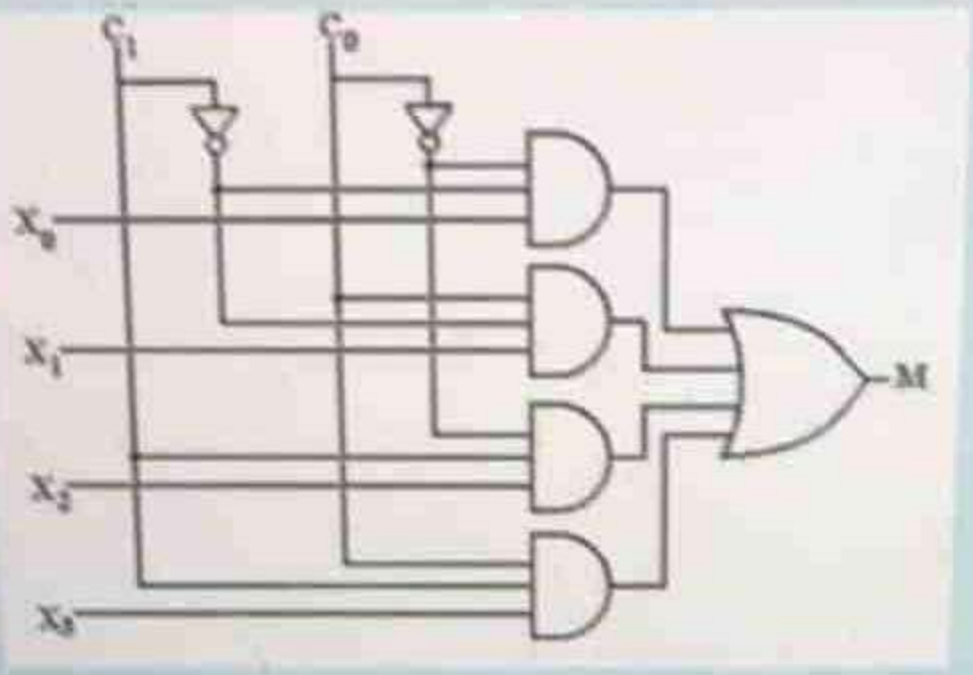
The binary numbers A = 1100 and B = 1001 are applied to the inputs of a comparator. What are the output levels?

Select one:

- a. A > B = 1, A < B = 0, A < B = 1
- b. A > B = 0, A < B = 1, A = B = 1
- c. A > B = 1, A < B = 0, A = B = 0
- d. A > B = 0, A < B = 1, A = B = 0



In the given 4-to-1 multiplexer, if c1 = 0 and c0 = 1 then the output M is _____



Select one:

- a. X2
- b. X1
- c. X0
- d. X3

B

How is a J-K flip-flop made to toggle?

Select one:

- a. J = 0, K = 0
- b. J = 1, K = 1
- c. J = 0, K = 1
- d. J = 1, K = 0



What is the minimum number of 2 input NAND gates required to implement the function $F = (x'+y')(z+w)$

Select one:

- a. 5
- b. 3
- c. 4
- d. 6

A

T-flip-flop is commonly used as

Select one:

- a. a digital counter only
- b. none of the above
- c. a delay switch
- d. a digital counter and frequency divider



If an active-HIGH S-R latch has a 0 on the S input and a 1 on the R input and then the R input goes to 0, the latch will be _____.

Select one:

a. hold

b. SET

c. RESET

d. invalid



Which one of the following set of gates are best suited for 'parity' checking and 'parity' generation.

Select one:

- a. NAND gates
- b. EX-NOR or EX-OR gates
- c. NOR gates
- d. AND, OR, NOT gates



Find the minimal form of the k map shown below?

cd \ ab	00	01	11	10
00	1	X	X	1
01	X			1
11				
10	1			X

Select one:

a. b'd' + b'c'

b. b'd' + b'c' + c'd'

c. b'd'

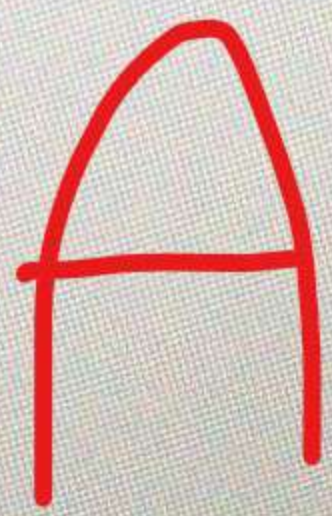
d. b'd' + a'b'c'd'

C

ESIGN-3

The simplest form for Boolean function ?

$F(A,B,C) = \sum m(0,1,3,4,5)$?



Select one:

- a. $F(A,B,C) = \overline{A}C + B$
- b. $F(A,B,C) = \overline{A} + \overline{B} + \overline{C}$
- c. $F(A,B,C) = \overline{A} + \overline{B}.C$
- d. D). $F(A,B,C) = \overline{A}\overline{B} + \overline{C}$

Minimize the following boolean function :

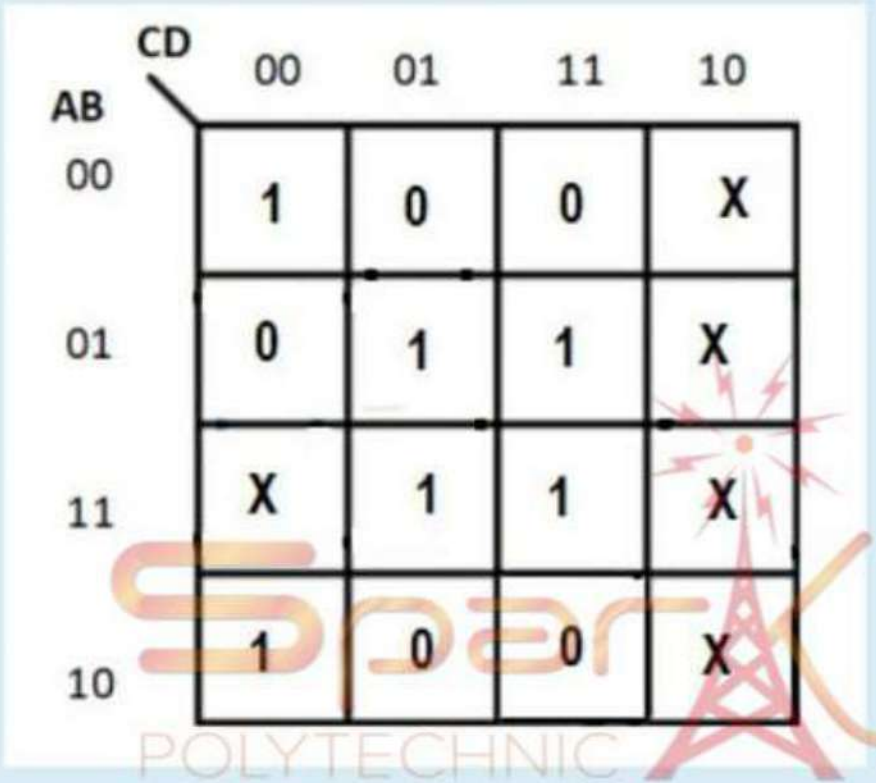
$$F=WZ'+WX'+WY'+WX'Y+WXY$$

Select one:

- a. W
- b. W ' + Y
- c. 1
- d. W ' + Z '

A

Find the minimal form of the k map shown below as POS ?



AB \ CD	00	01	11	10
00	1	0	0	X
01	0	1	1	X
11	X	1	1	X
10	1	0	0	X

Spark POLYTECHNIC

Select one:

- a. (B+D')(A+B'+C+D)
- b. (B+D')(B'+C+D)
- c. (B+D')(B'+D)
- d. 0

C

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The following boolean function can be simplified to :

WX'YZ' + WX'YZ + WXYZ' + WXYZ

Select one:

- a. XYZ
- b. W+Y
- c. W Y
- d. Y

C

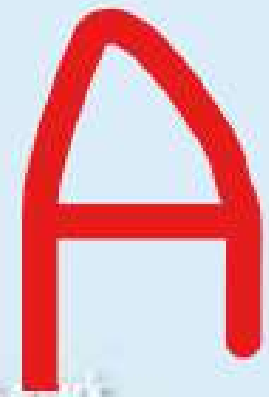
Spark POLYTECHNIC

The POS function which equivalent to SOP function :

$$F(A,B,C) = ABC + AB'C + A'B'C + AB'C' + A'BC$$

Select one:

- a. (A + B + C)(A + B' + C)(A' + B' + C)
- b. (A' + B')(A + B + C)(B' + C)
- c. 1
- d. 0



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The HEXADECIMAL number $(3FC.CC)_{16}$ is equivalent to ..... in OCTAL :

Select one:

- a. 1774.344
- b. 7760.344
- c. 7760.63
- d. 1774.63

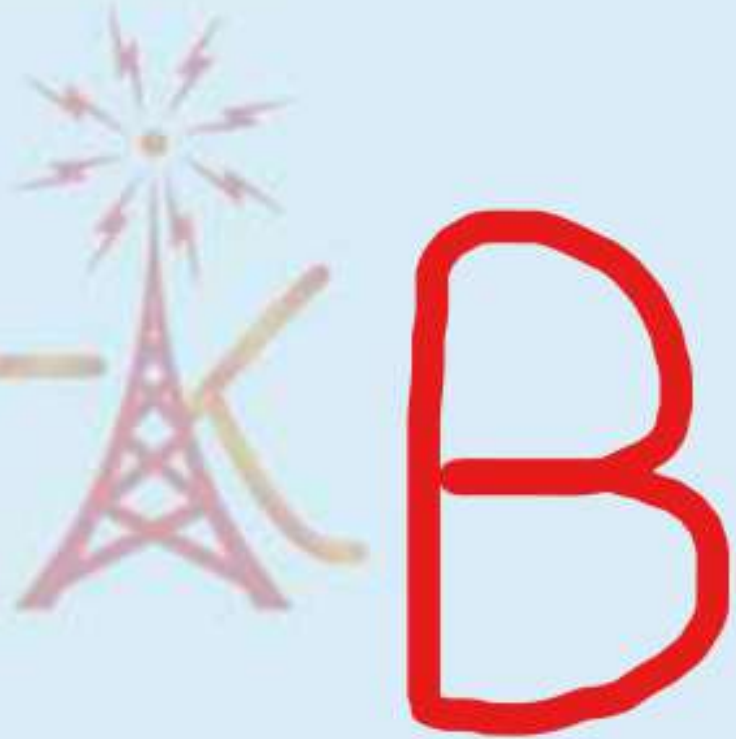
D

Spark POLYTECHNIC

If A and B are the inputs of a half adder, the sum is given by _____

Select one:

- a. A XNOR B
- b. A XOR B
- c. A AND B
- d. A OR B



Find the F' as SOP for the function F = A'B'C +A'BC' + ABC + A'B'C' + ABC'

Select one:

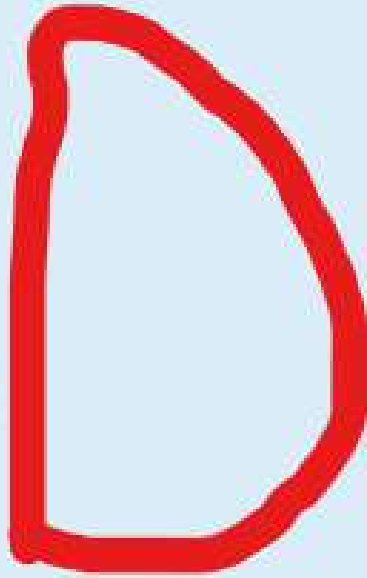
- a. A'B'C' +A'BC +A'BC'
- b. A'BC + AB'C' +AB'C
- c. A'B'C' +A'B'C +A'BC'+ABC'+ABC
- d. ABC+AB'C+A'B'C



The result of subtraction $(DFB)_{16}$ - $(1767)_8$ in octal is:

Select one:

- a. $(5106)_{octal}$
- b. $(5006)_{octal}$
- c. $(5774)_{octal}$
- d. $(5004)_{octal}$



What is the simplest form for Boolean function ?

F(A,B,C)= A'B' +AB' +A'BC

Select one:

- a. A'B'C'
- b. B' + A'C
- c. A + B + C
- d. A' + B'C



Spark POLYTECHNIC
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Minimize the function F and express it in SOP:

$$F(A,B,C,D) = \sum(0,1,2,3,4,5,6,7,10,11,12,13,14,15)$$

Select one:

- a. F=A+B'+C'D
- b. F=A'+B+C
- c. F=ABC+A'BCD+BCD'
- d. F=A'B+C'D+A

B

What is the minimum number of NAND gates (only) can be used to design the function **F=A'B + AB'**

Select one:

- a. 5
- b. 6
- c. 4
- d. 3

C

Spark POLYTECHNIC

The binary code $(11110101)_{gray}$ is equivalent to $(\quad)_{10}$

Select one:

- a. 198
- b. 152
- c. 166
- d. 150

Spark POLYTECHNIC

C

The result of -26 + -28 in 2's complement is :

Select one:

- a. 00110110
- b. 11001010
- c. 11001001
- d. 10101111



You have two types of DMUX 1-to-4 and 1-to-8. How many DMUX are needed from type 1 and tpe 2 to build 1-to-32 DMUX

Select one:

- a. 1 5
- b. 1 3
- c. 1 8
- d. 1 4



The number of full and half-adders required to add 16-bit numbers is

Select one:

- a. 1 half-adder, 15 full-adders
- b. 4 half-adders, 12 full-adders
- c. 8 half-adders, 8 full-adders
- d. 16 half-adders, 0 full-adders



The representation of octal number (532.2)8 in decimal is _____

Select one:

- a. (346.25)10
- b. (340.67)10
- c. (532.864)10
- d. (531.668)10



With regard to a D latch, _____.

C

Select one:

- a. the Q output is HIGH regardless of EN's input state
- b. the Q output is opposite the D input when EN is LOW
- c. the Q output follows the D input when EN is HIGH
- d. the Q output follows the D input when EN is LOW

Which of the following expressions is not equivalent to X ' ?

Select one:

- a. X NAND X
- b. X NOR 1
- c. X NOR X
- d. X NAND 1



The digital multiplexer is basically a combination logic circuit to perform the operation

Select one:

- a. AND-OR
- b. OR-AND
- c. AND-AND
- d. OR-OR



The decimal equivalent of the binary number (1011.011)2 is _____

Select one:

- a. (11.175)10
- b. (10.124)10
- c. (11.375)10
- d. (10.123)10

